2013 HCI H1 Chemistry Preliminary Examination Answer Scheme

Paper 1

1	**B**	6	**B**	11	**A**	16	**C**	21	**C**	26	**D**
2	**C**	7	**A**	12	**C**	17	**B**	22	**C**	27	**B**
3	**D**	8	**A**	13	**D**	18	**C**	23	**B**	28	**A**
4	**D**	9	**C**	14	**B**	19	**A**	24	**D**	29	**B**
5	**B**	10	**A**	15	**A**	20	**A**	25	**D**	30	**C**

A = 8; B = 8; C = 8; D = 6

Paper 2 (Section A)

1 a) i) Si or P

ii) $2Al(s) + 3Cl_2(g) \rightarrow Al_2Cl_6(s)$ or $2AlCl_3(s)$

(iii) argon has complete/ full outer shell of electrons or octet or IE too high

b)
$NaCl(s) + aq \rightarrow Na^+(aq) + Cl^-(aq)$

$AlCl_3(s) + aq \rightarrow Al^{3+}(aq)$ or $[Al(H_2O)_6]^{3+} + 3Cl^-(aq)$
$[Al(H_2O)_6]^{3+} + H_2O \rightleftharpoons [Al(H_2O)_5OH]^{2+} + H_3O^+$

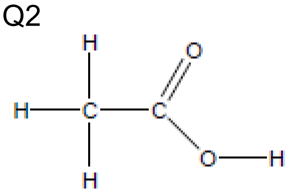
$SiCl_4(l) + 4H_2O \rightarrow SiO_2.2H_2O(s) + 4HCl(aq)$

c)i)
nitrogen atom 1 lone pair

sulphur atom 2 lone pairs

ii) angle a or S because lone pair- lone pair repulsions are stronger than lone pair- bond pair repulsions

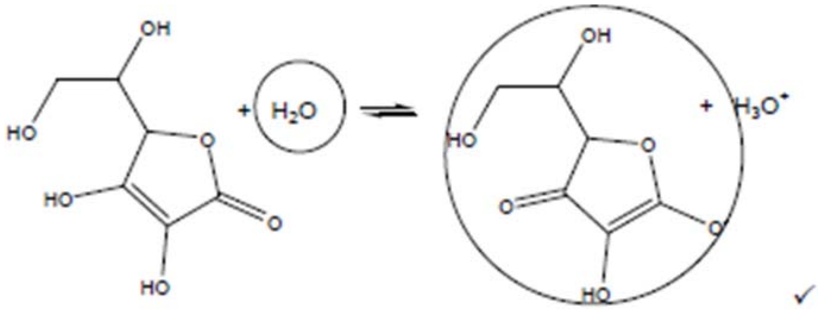
Q2



b) $K_2Cr_2O_7(aq)$ + (acidified) /with $H_2SO_4(aq)$
heat under reflux

c) ethanal
reasoning: absorption at 1720-1740 indicate a carbonyl compound;
no broad peak at 25000-3200 suggest absence of –OH found in acid or alcohols.

d) i) a proton/ H^+ acceptor



ii) ALLOW If only $-O^-$ is circled

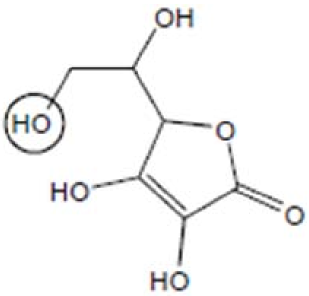
iii) gas is CO_2

$(C_6H_7O_6)_2^-\ Ca^{2+}$ or $(C_6H_7O_6)_2Ca$

e)
restricted rotation around C=C bond

the two –OH groups in E300 can only be on the same side of the C=C because the ring structure will not allow them to be on opposites /cannot rotate.

f)



ii) –OCO- or -OOC-

iii) Concentrated sulfuric acid

3 a) i) rate of forward reaction is equal to rate of backward

ii)

$Kc = [N_2O_4]/[NO_2]^2$

iii) $Kc = (0.400/2) / \{(3.2)/2\}^2 = (0.200)/(1.6)^2 = 7.81 \times 10^{-2}\ dm^3\ mol^{-1}$

b) concentration of N_2O_4 will decrease and conc. of NO_2 will increase

4. a) alkane → ester → alcohol

b (i) $C_3H_6O_2 + 7/2\ O_2 \rightarrow 3CO_2 + 3H_2O$ or equation multiplied through by 2.

(ii) Enthalpy change when **1 mol** of substance is formed **from its elements.**
1 mark for each of the points in bold
Mention of standard states or under standard conditions or 1 bar pressure.

(iii) $\Delta_f H^{\ominus}(C_3H_6O_2) = \mathbf{3}\Delta_c H^{\ominus}(C) + \mathbf{3}\ \Delta_c H^{\ominus}(H_2) - \Delta_c H^{\ominus}(C_3H_6O_2) =$

$= ((3\times-393.5) + (3\times-285.8) - (-1592.1))$ kJ mol^{-1} = **$-$ 445.8 kJ mol^{-1}**

Correctly multiplying carbon and hydrogen values by 3 or a 3:3:1 ratio of $C:H_2:C_3H_6O_2$ (1)

Correct signs, i.e. $-\Delta_c H^{\ominus}$(ester) + $\Delta_c H^{\ominus}$(elements)
Correct final answer to 1 d.p. (1)
Allow ecf from an earlier penalised error if it has been worked through correctly.

c) Thermal energy added to water = 4.18 J K^{-1} g^{-1} × 10.0 K × 300 g = 12540 J (1)
Thermal energy added to copper = 0.384 J K^{-1} g^{-1} × 10.0 K × 250 g = 960 J (1)
Total energy = **13.5 kJ** (3 s.f. required) (1)

Answer must be in kJ, not Joules, but no penalty for omitting to write kJ.

d)

Amount of ester = 0.980 g / 74.0 g mol^{-1} = 0.0132 mol (1)
Theoretical energy released = 0.0132 mol × 1592.1 kJ mol^{-1} = 21.1 kJ (1) 3 s.f.
Allow ecf with amount of ester.

e) The methyl ethanoate will be easier to light (more volatile)

Paper 2 (Section B)

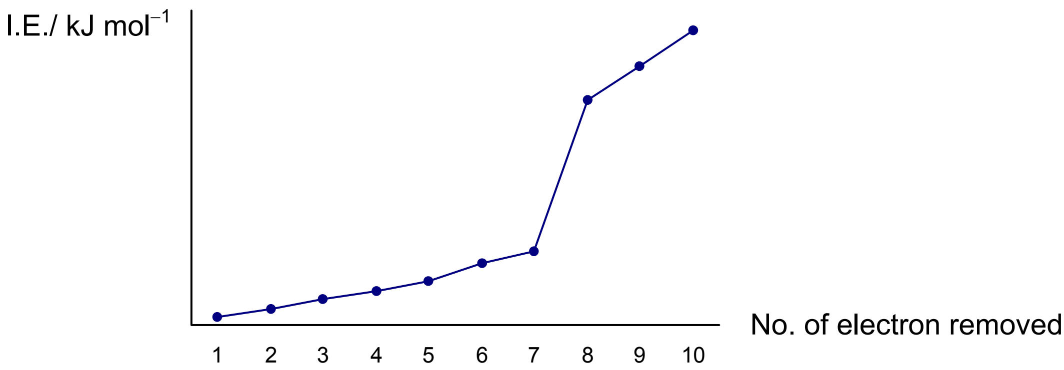
1 **(a)** **(i)** $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ [1]

(ii) Isotopes are atoms of the same element that has the same number of protons but different number of neutrons / or different mass number. [1]

(iii) 17 protons, 20 neutrons and 17 electrons [1]

(iv) A_r of C*l* = $\dfrac{(1\times37)+(3\times35)}{4}$ = **35.5** (no units) [1]

(v)



Correct axes (units optional) and shape [1]

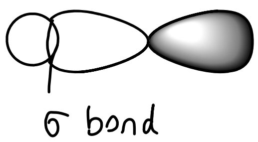
Successive I.E. increase as the remaining electrons are closer to the nucleus due to the decreasing number of electrons being attracted by the same number of protons / or increasing effective nuclear charge. [1]

There is a large increase from the 7^{th} to 8^{th} I.E. as the 8^{th} electron is removed from the inner quantum shell which is much closer to and less shielded from the nucleus. [1]

(b) **(i)** Oxidation number increases from 0 in Cl_2 to +1 in NaOC*l* and decreases from 0 to –1 in NaC*l*. [1]

Cl_2 undergoes disproportionation / both oxidation and reduction. [1]

(ii) [1]



The H–C*l* bond consists of a σ bond formed by head on overlap of s orbital of H with p orbital of C*l*. [1]

(c) **(i)** $[H^+]$ in HC*l* = $10^{-1.3}$ = **0.0501 mol dm^{-3}** [1]
$[H^+]$ in CH_3COOH = $10^{-3.0}$ = **0.001 mol dm^{-3}**

$[H^+]$ in CH_3COOH is much lower as CH_3COOH is a weak acid which dissociates partially while HC*l* is a strong acid which dissociates completely.

Accept explanation in terms of strength of HCl [1]

(ii) $2\ CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$

Accept ionic equation [1]

(iii) η_{H+} available = 0.0501 x 20/1000 = 1.002 x 10^{-3} mol

Volume of H_2 = ½ x η_{H+} x 24000 = ½ x 0.001002 x 24000 = **12.0 cm³** [1]

(iv) H_2 is evolved more slowly due to lower $[H^+]$ in CH_3COOH (since rate α $[H^+]$). [1]

The equilibrium $CH_3COOH \rightleftharpoons CH_3COO^- + H^+$ continually shifts to the right as H^+ is used up. [1]
The same volume of H_2 is produced since the amount of H^+ ions available are the same in both acids.

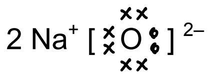
(d) An acidic buffer consists of a weak acid and its conjugate base and is able to resist pH changes when small amount of acid or base is added to it. [1]

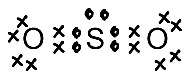
The buffer in blood contains H_2CO_3 with HCO_3^-. Excess H^+ and OH^- ions are removed as follows.

$HCO_3^- + H^+ \rightarrow CO_2 + H_2O$ [1]
or $HCO_3^- + H_3O^+ \rightarrow CO_2 + 2H_2O$
or $HCO_3^- + H_3O^+ \rightarrow H_2CO_3 + H_2O$

$H_2CO_3 + OH^- \rightarrow HCO_3^- + H_2O$ [1]

2 (a) (i)



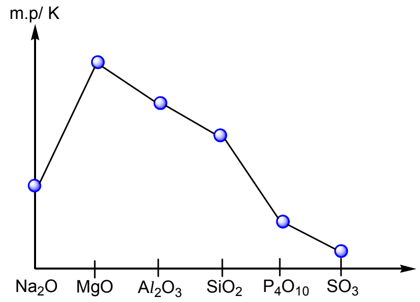


2x[1]

(ii) Na_2O has a <u>giant ionic</u> structure which consists of Na^+ and O^{2-} ions held by <u>strong ionic bonds</u>. Hence, Na_2O has a <u>high</u> melting point. [1]

SO_2 has a <u>simple covalent</u> structure where SO_2 molecules are held by <u>weak pd–pd attractions</u>. Hence, SO_2 has a <u>low</u> melting point. [1]

(iii)



Correct axes (units optional) and shape [1]
Accept SO_2 or SO_3

(iv) Na_2O(s) – solution turns <u>blue</u> as a strongly <u>alkaline</u> solution is formed. [1]

$Na_2O + H_2O \rightarrow 2NaOH(aq)$

Accept ionic equation [1]

Al_2O_3(s) – solution remains <u>purple</u> as solid is <u>insoluble</u> in water. [1]

(b) (i)

CH_2—$CH_2CH_2CH_3$
|
OH

or

H_3C—CH—CH_2OH
|
CH_3

[1]

(ii) $H_2C{=}CHCH_2CH_3$ or H_3C—C(H)=C(H)—CH_3 [1]

(iii) $K_2Cr_2O_7$, dilute H_2SO_4, heat [1]
A – orange solution turns green; **B** – solution remains orange [1]

or

$KMnO_4$, dilute H_2SO_4, heat

A – purple solution turns colourless; **B** – solution remains purple or

I_2(aq), NaOH(aq), warm
A – yellow crystals/ ppt; **B** – no ppt

(c) (i) Type of reaction: (nucleophilic) substitution [1]
Reagents: PCl_5 , (r.t.) / $SOCl_2$, warm / conc. HC*l*, $ZnCl_2$

[1]

(ii) $CH_3CH{=}CH_2 + HCl \rightarrow CH_3CHClCH_3$

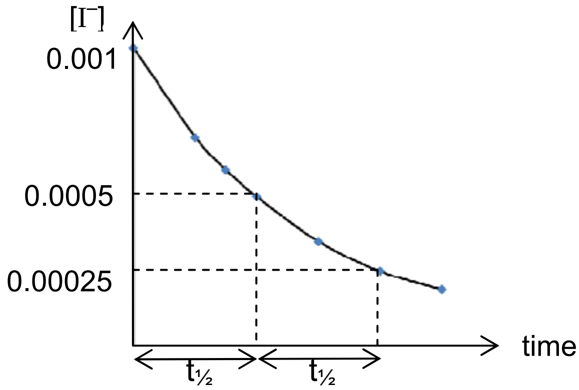
ΔH_r = [BE(C=C) + BE(H–C*l*)] – [BE(C–C) + BE(C–H) + BE(C–C*l*)]
= (610 + 431) – (350 + 410 + 340) 2x[1]
= 1041 – 1100
= **–59 kJ mol^{-1}** (sign and units) [1]

(d) BE(C–C*l*) = 340 kJ mol^{-1}
BE(C–Br) = 280 kJ mol^{-1}
BE(C–I) = 240 kJ mol^{-1} [1]

The <u>rate</u> of hydrolysis <u>increases</u> as the <u>C–X</u> bond gets <u>weaker</u>. [1]

Hence, <u>CH_3CHICH_3</u> hydrolyses <u>most rapidly</u> to give iodide ions which form a ppt with Ag^+ immediately (while $CH_3CHClCH_3$ hydrolyses least rapidly hence ppt is observed only after 10 minutes). [1]
or
<u>Rate</u> of hydrolysis <u>increases</u> from <u>$CH_3CHClCH_3$ to $CH_3CHBrCH_3$ to CH_3CHICH_3</u> and hence a ppt is observed immediately for CH_3CHICH_3.

3 (a) (i)



Correct axes and shape [1]
Shows $t_{½}$ is constant [1]

(ii) Compare Expt 1 and 2:
As $[H_2O_2]$ increases by 0.07/0.05 = 1.4, so does rate
$\Rightarrow$ order w.r.t. $[H_2O_2]$ = **1**
or

$$\left(\frac{0.07}{0.05}\right)^n = \frac{1.4}{1.0} \Rightarrow \text{order w.r.t. } [H_2O_2] = n = \mathbf{1}$$ [1]

Compare Expt 1 and 3:
As $[H_2O_2]$ increases by 0.09/0.05 = 1.8 and $[H^+]$ increases by 0.07/0.05 = 1.4, rate increases by 1.8 times.
$\Rightarrow$ rate is independent of $[H^+]$ / order w.r.t. $[H^+]$ = **0**
or

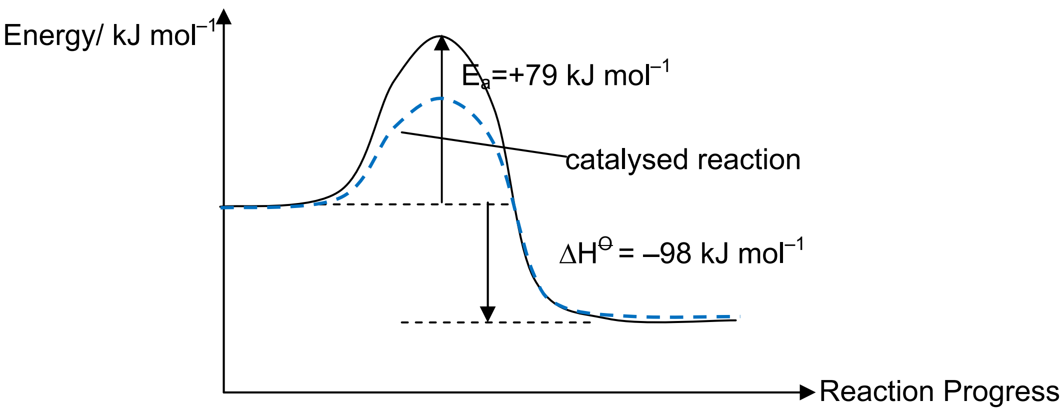
$$\left(\frac{0.07}{0.05}\right)^m \left(\frac{0.09}{0.05}\right) = \frac{1.8}{1.0} = 1$$

$\Rightarrow$ order w.r.t. $[H^+]$ = m = **0** [1]

If both orders are correct but no working/explanation given – award [1]

(iii) Rate = $k\,[H_2O_2]\,[I^-]$ [1]
Units for k = $mol^{-1}\,dm^3\,s^{-1}$ or $mol^{-1}\,dm^3\,min^{-1}$ [1]
*ecf from **(a)(ii)***

(b) (i)



Correct axes and shape [1]
Correct labels for ΔH and E_a [1]

(ii) *Correct profile for catalysed reaction* [1]

A catalyst provides a <u>different reaction pathway</u> with <u>lower</u> activation energy. [1]
The <u>proportion</u> of molecules with energy greater or equal to activation energy increases.
<u>Number of effective collision</u> increases and reaction rate increases.

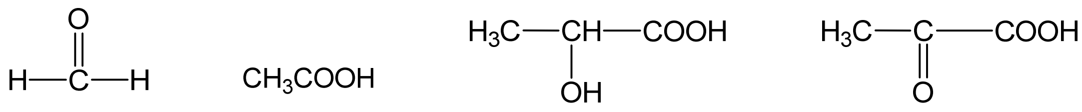
(iii) The rate constant will <u>increase</u>. [1]

From the rate equation, if rate increases despite all concentrations remaining the same, then the value of *k* must have increased. [1]

or

Using Arrhenius equation, $k = A\ e^{-Ea/RT}$, if activation energy, E_a, decreases, k will increase.

(c) 4x[1]



C **D** **E** **F**

C gives a Ag mirror with Tollens' reagent ⇒ **C** is an <u>aldehyde</u> [1]

D and **E** contains –<u>COOH</u> group which reacts with Na_2CO_3(aq) to give <u>CO_2</u> gas. [1]

E undergoes <u>oxidation</u> with acidified $K_2Cr_2O_7$ to give **F**

F undergoes <u>condensation</u> with 2,4–DNPH to give an orange–yellow ppt
⇒ **F** contains a <u>ketone</u> ⇒ **E** contains a <u>2° alcohol</u>

***F** contains ketone + condensation* [1]
***E** contains 2° alcohol + oxidation* [1]
or award 1 mark for any 2 of the 4 points